

SHEET 1 - COVER - DESIGN BRIEF, DESIGN VIEWS, SHEET INDEX

DESIGN BRIEF (client, 2026-10-01)

Premium interchangeable knitting-needle case in genuine or faux leather: structured construction, finished edge stitching, folding side flaps, top flap, and a leather strap with metal hardware that wraps the whole case. Minimal and premium in appearance - not a fabric organiser. The case opens fully.

- Snaps hold the front panel to the two folded side flaps.
- The strap closes everything: buckle and keeper are visual, the strap tongue snaps to a stud on the front panel. Client logo embossed on the top flap.

STORAGE SURFACES - front to back

Front panel > Panel 1 > Panel 2 > Panel 3 > Back wall (Panel 4)

- Panels 1-3 are separate leather pages, permanently sewn into the bottom of the case, evenly spaced, and able to flip individually like book pages. They are NOT removable.
- Panel 4 is the inside back wall of the case itself: the needle loops are sewn / threaded directly into the back-wall lining. It also carries the large snap pocket for the cables.
- Panel 2 carries two small snap pockets for the accessories.
- Every pair of tips has its own elastic loop, threaded in and out of the panel, with the needle size marked above the pair.

CONTENTS TO BE HOUSED

- 34 x 10 cm tips (17 pairs, 2.0-10 mm) on panels 1 and 2
- 30 x 5 cm tips (15 pairs, 2.0-8 mm) on panels 3 and 4
- 5 cables 25 / 40 / 60 / 80 / 100 cm back-wall pocket (panel 4)
- 6 end caps, 4 keys, 3 connectors, panel 2 pockets
- 1 leather grip patch (replaces the rubber grip disc)

CLOSED SIZE 21.0 x 13.0 x 6.0 cm (W x H x D, from the cardboard prototype)

MATERIALS

Outer: full-grain leather 1.2-1.4 mm, tan (or PU faux leather 1.0-1.2 mm). Strap: 1.4 mm, dark brown. Lining and pages: 0.6-1.0 mm matching leather. Hardware: antique brass - 30 mm buckle (decorative), 12.5 mm and 10 mm snaps. Elastic: 15 mm knit, colour-matched. Thread: bonded polyester, contrast.

LOGO (client artwork, right) blind / heat emboss 2.5 x 2.5 on the top flap,

centred 3.0 from the right edge and 8.0 below the top of the closed case (sheet 2). Vector artwork to be supplied for the emboss die; raster shown for placement.

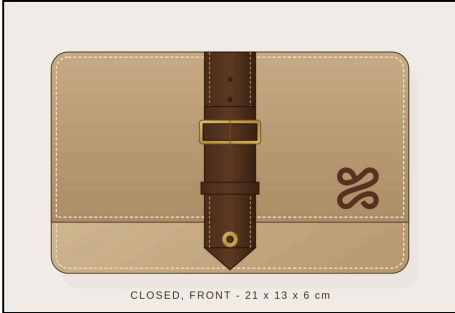
TOP FLAP 10.0 deep, as in the reference photos (client decision); the cardboard

prototype measured 5.0. Pockets sized to the maximum that leaves the needle zones intact: cable pocket 9.7 x 9.0, accessory pockets 3.9 x 4.1 (sheets 5, 7).

SHEET INDEX

- 1 Cover (this sheet)
- 2 Outer shell - flat pattern, outer face
- 3 Assembly - closed views, section, open layout, closing sequence
- 4 Panel 1 (page): 10 cm tips 2.0-5.0 mm
- 5 Panel 2 (page): 10 cm tips 5.5-10 mm + two accessory pockets
- 6 Panel 3 (page): 5 cm tips 2.0-5.0 mm
- 7 Panel 4 (back wall): 5 cm tips 5.5-8 mm + cable pocket
- 8 Details: strap, elastic threading, page hinge, pockets
- 9 Contents checklist, bill of materials, construction

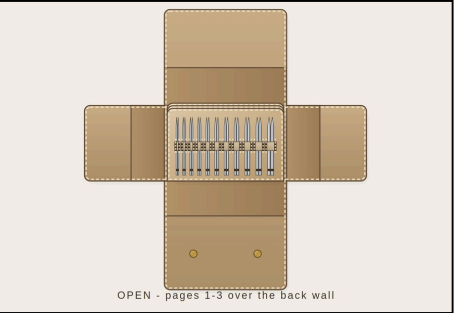
DESIGN VIEWS - generated from these drawings (render_views.py); not photographs



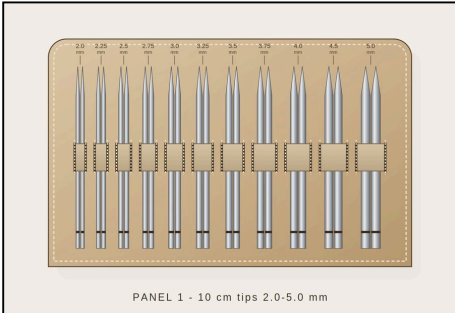
V1 Closed, front: strap, snap, logo on flap



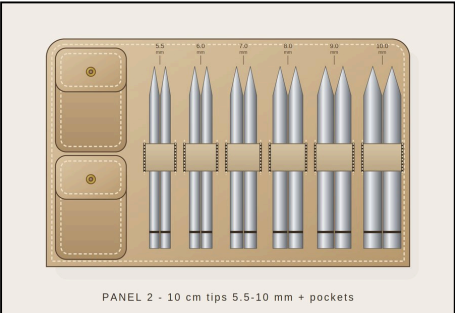
V2 Closed: strap wraps over the top



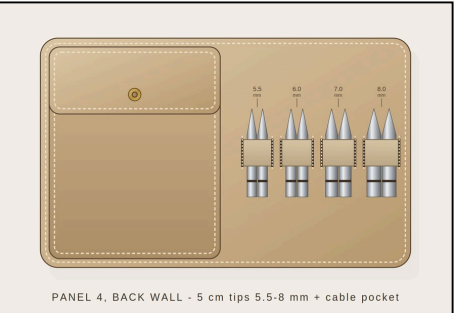
V3 Open: pages 1-3 over the back wall



V4 Panel 1: 10 cm tips 2.0-5.0 mm



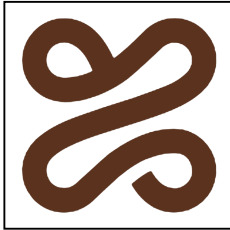
V5 Panel 2: 10 cm tips 5.5-10 mm + pockets



V6 Panel 4, back wall: 5 cm tips 5.5-8 + cables

HOW THE DRAWINGS RELATE TO THE BRIEF

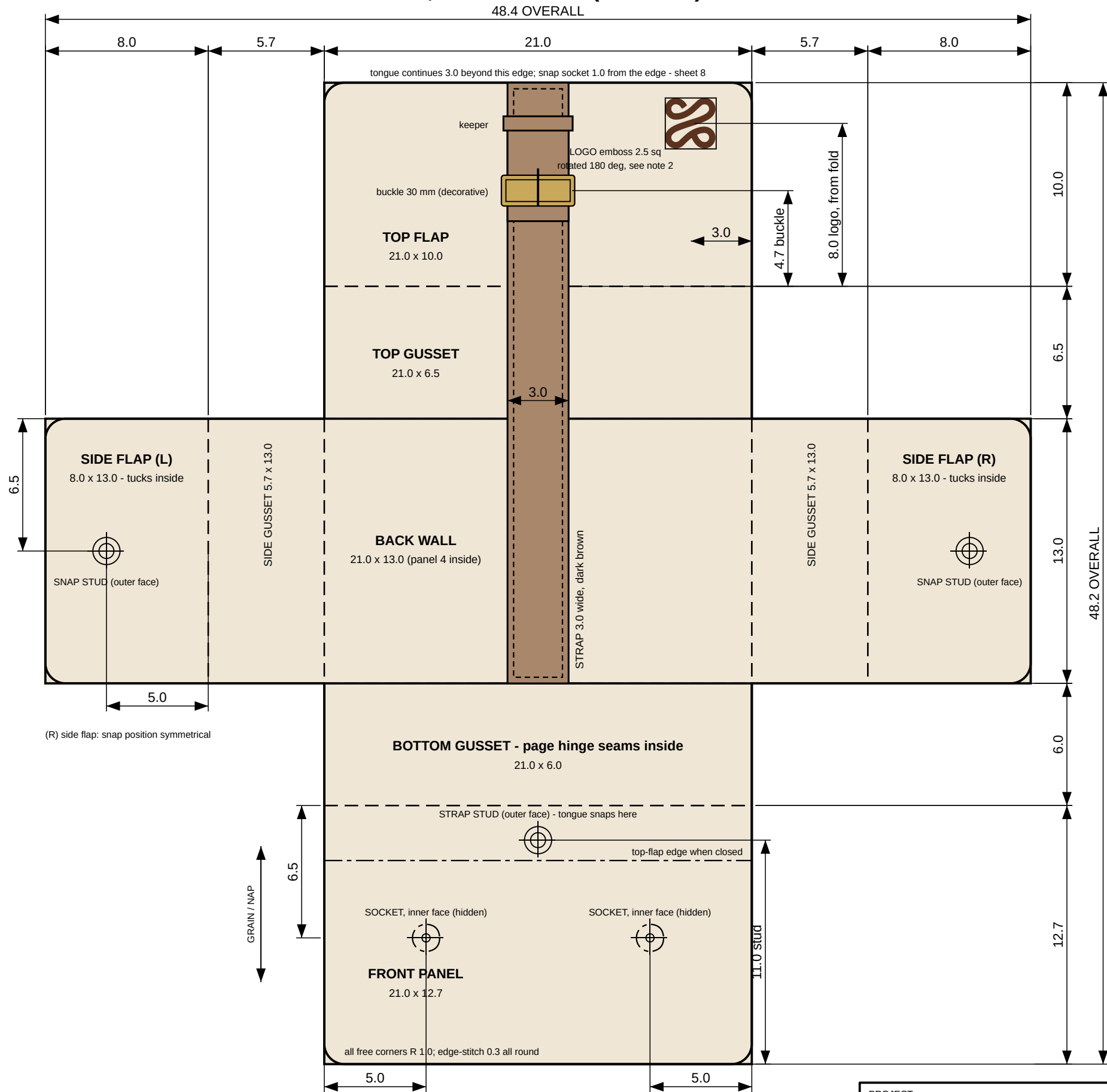
- Dimensions come from the client's cardboard prototype (21.0 x 13.0 base, 6.0 gussets, 8.0 side flaps, loop widths 0.6-2.2) - see the record on sheet 9.
- Construction follows the client's reference photos (kept in reference/) and brief; where photos and prototype differed (flap depth) the client chose 10.0.
- The views above are drawn from the same geometry as sheets 2-8, so they show the case as specified: logo upright on the closed flap, pockets at final size.
- Sheets 4-7 are full size (1:1): print at 100 % for slit-cutting templates.
- Open questions for the factory / client are listed on sheet 3.



upright as shown when the case is closed

PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Cover - design brief, design views, sheet index	SCALE -	UNITS cm (page A3)
SOURCE Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	DATE / REV 2026-10-01 Rev E	SHEET 1 of 9

SHEET 2 - OUTER SHELL - FLAT PATTERN, OUTER FACE (DIE LINE)



LEGEND

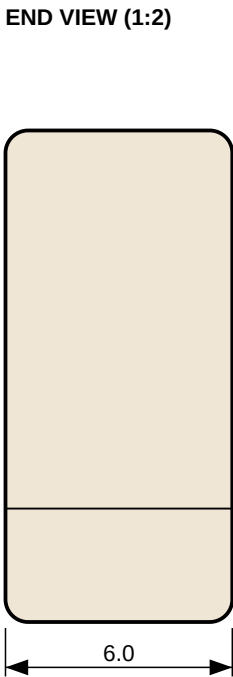
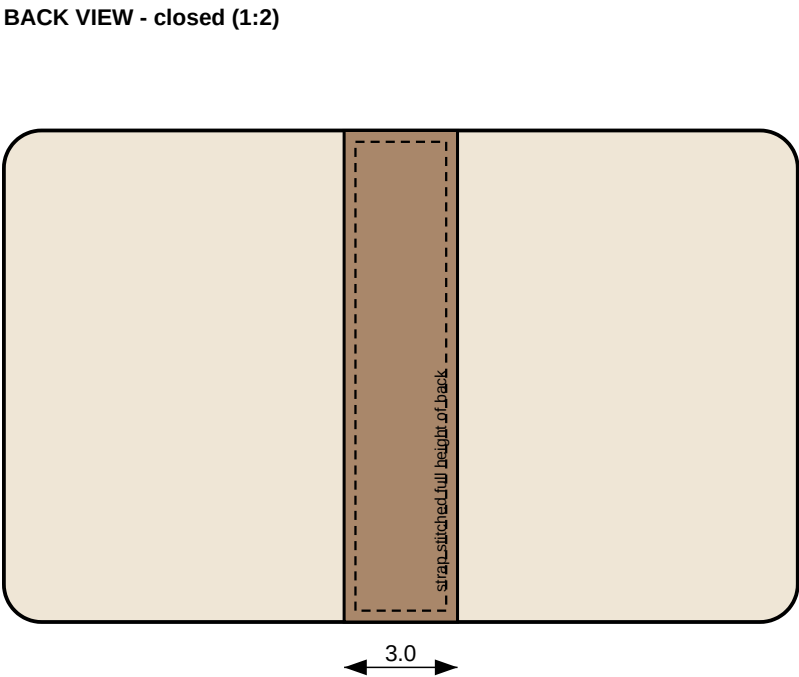
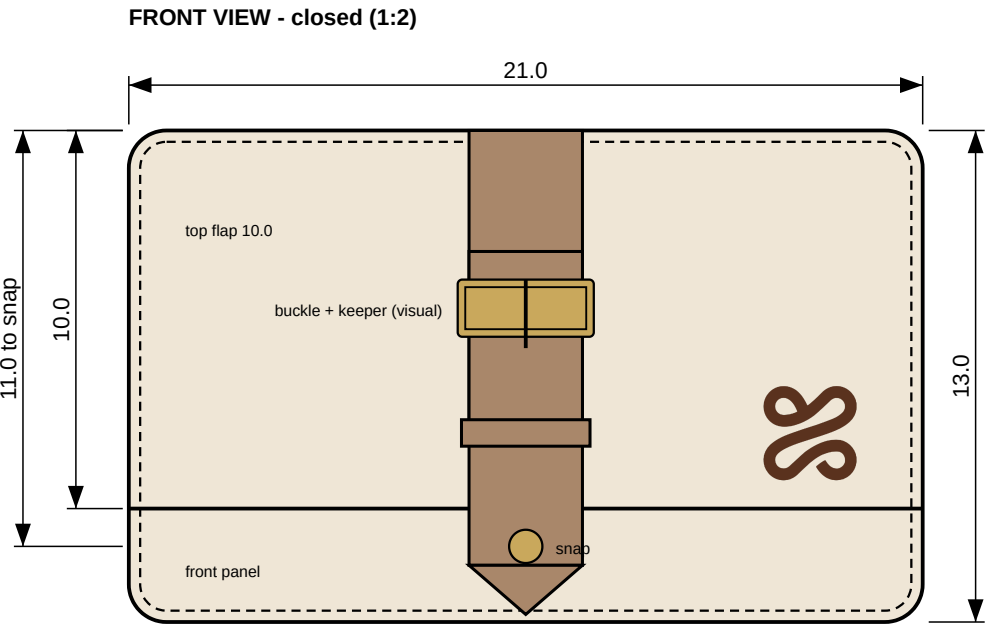
- Cut line (net finished size)
- Fold / score line
- Hidden (far face / when closed)
- Stitch line
- Snap on this face
- Snap on far face

NOTES

- All dimensions in cm, finished (net) sizes as measured on the cardboard prototype. No seam allowance included: add 0.8 for turned seams, 0 for bound / edge-painted edges (reference shows edge-stitched, turned edges).
- Drawn with the OUTER face up (strap, buckle, logo side). Pattern assumed symmetrical about its vertical centre line; left side measured equal to the right (5.7 / 8.0). The top flap folds over the top and down the front, so its buckle, keeper and logo are drawn rotated 180 deg and dimensioned from the fold line (= top of the closed case).
- Construction: trifold clutch (reference photos). Side flaps fold in and tuck inside, front panel folds up over them and snaps to the side flaps, top flap folds down, strap tongue snaps to the front panel. Pages 1-3 are sewn into the bottom gusset; the back wall lining is storage panel 4 (sheet 7).
- Gusset depths as measured: 5.7 (sides), 6.0 (bottom), 6.5 (top). Keep as measured (Q1, sheet 3).
- Strap: stitched to the back wall, top gusset and top flap; 30 mm buckle and keeper on the flap are visual only. The tongue (3.0 beyond the flap) snaps to a stud on the front panel 11.0 below the top. Side-flap snaps: 12.5 mm, studs outer face, sockets front-panel inner face - they hold the front panel to the side flaps.
- Outer: full-grain leather 1.2-1.4 mm (or PU 1.0-1.2), tan; strap dark brown. Board 1.0 mm in back wall and front panel; lining 0.6 mm, same tan (reference).
- Top flap 10.0 as in the reference photos (client, 2026-10-01); the prototype measured 5.0 - superseded.

PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Outer shell - flat pattern, outer face (die line)	1:2.2 (not to scale - use dims)	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	2026-10-01 Rev E	2 of 9

SHEET 3 - ASSEMBLY - CLOSED VIEWS, SECTION, OPEN LAYOUT, CLOSING SEQUENCE



OPENING / CLOSING

Open: unsnap the strap tongue, lift the top flap, fold the front panel down (its snaps release from the side flaps), fold the side flaps out. The case lies flat and the three pages flip like a book; the back wall is the fourth storage surface.

Close: side flaps in over the pages, front panel up and snap to the side flaps, top flap over, strap tongue snaps to the front.

STACK-HEIGHT CHECK (depth 6.0)

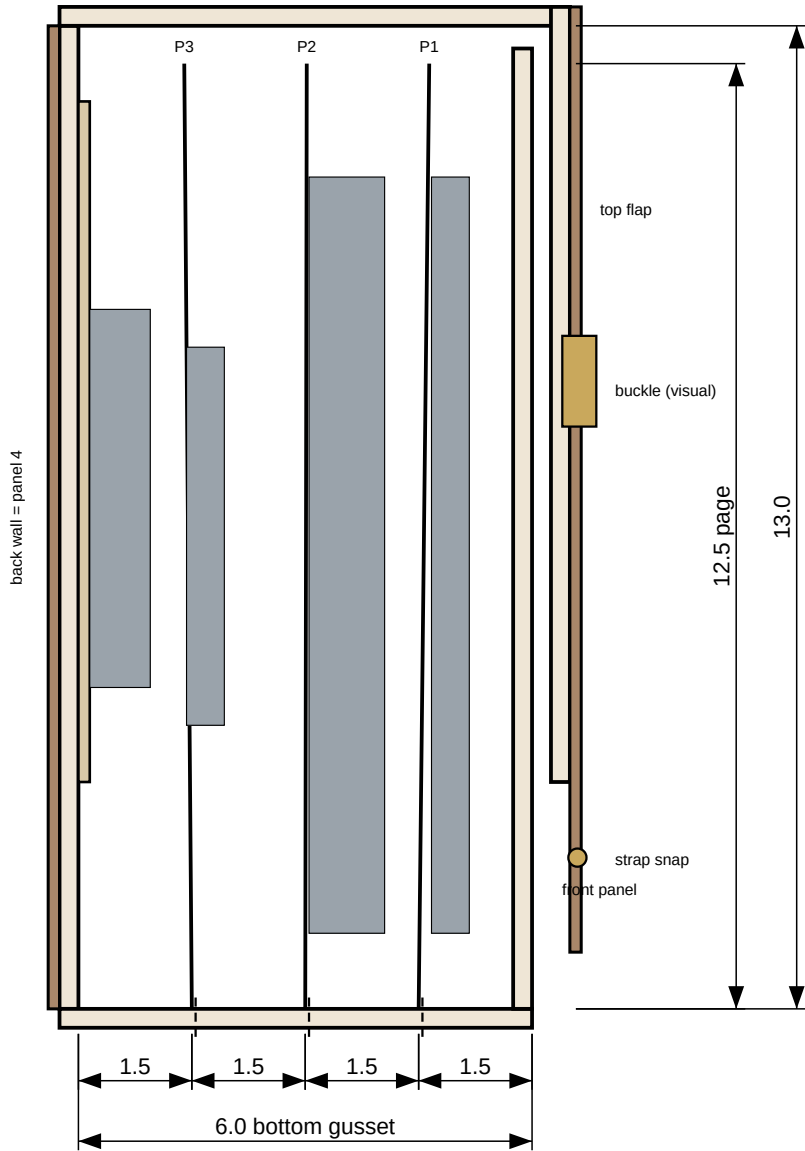
P4 back wall: 8 mm tips / cable pocket 1.3
P3 2 x 5 mm tips + page 0.7
P2 2 x 10 mm tips + page + pockets 1.3
P1 2 x 5 mm tips + page 0.7
front + back walls, side flaps 0.9
TOTAL 4.9 (OK)

OPEN QUESTIONS (confirm before cutting)

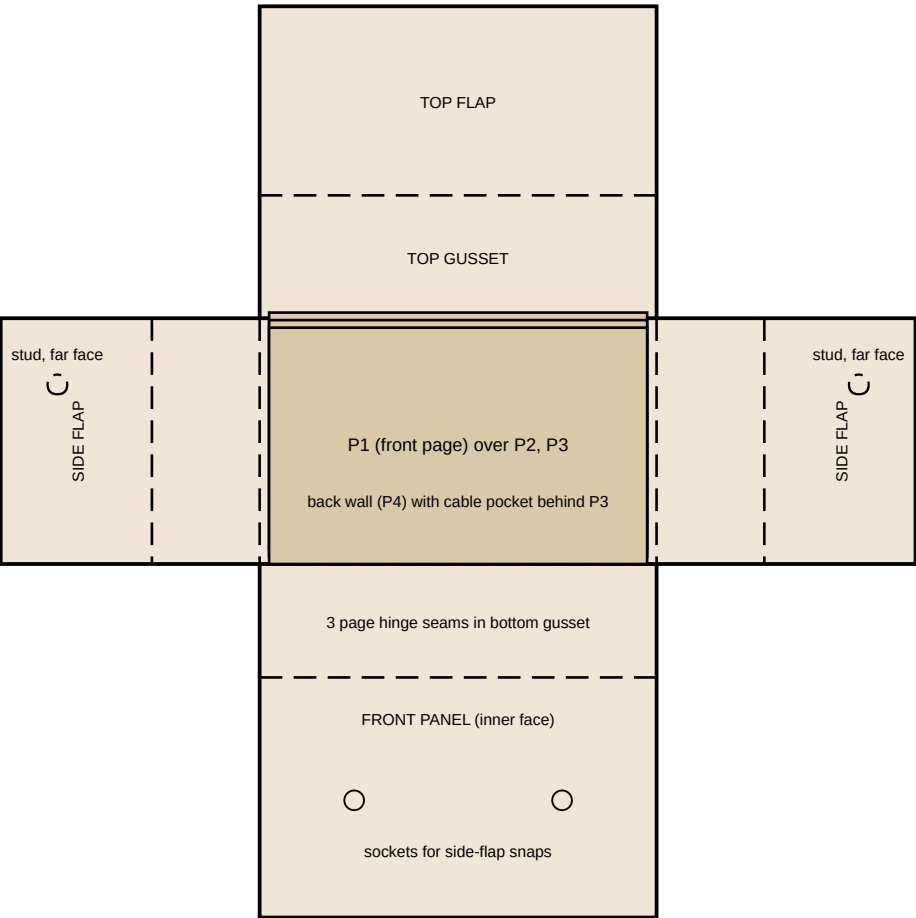
Q1 Keep gussets 5.7 / 6.0 / 6.5 as measured?
Q2 (closed) Top flap 10.0 per the reference photos, agreed 2026-10-01.
Q3 Side-flap snaps: spring snaps (drawn) or hidden magnetic snaps?
Q4 Genuine (full-grain 1.2-1.4) or PU faux leather for the shell?
Q5 Confirm real cap / key / connector / cable sizes against the pockets on sheets 5, 7.

SECTION A-A - vertical through the centre, closed (1:1)

Front to back: front panel > P1 > P2 > P3 > back wall (P4). Pages hinge on evenly spaced seams in the bottom gusset.

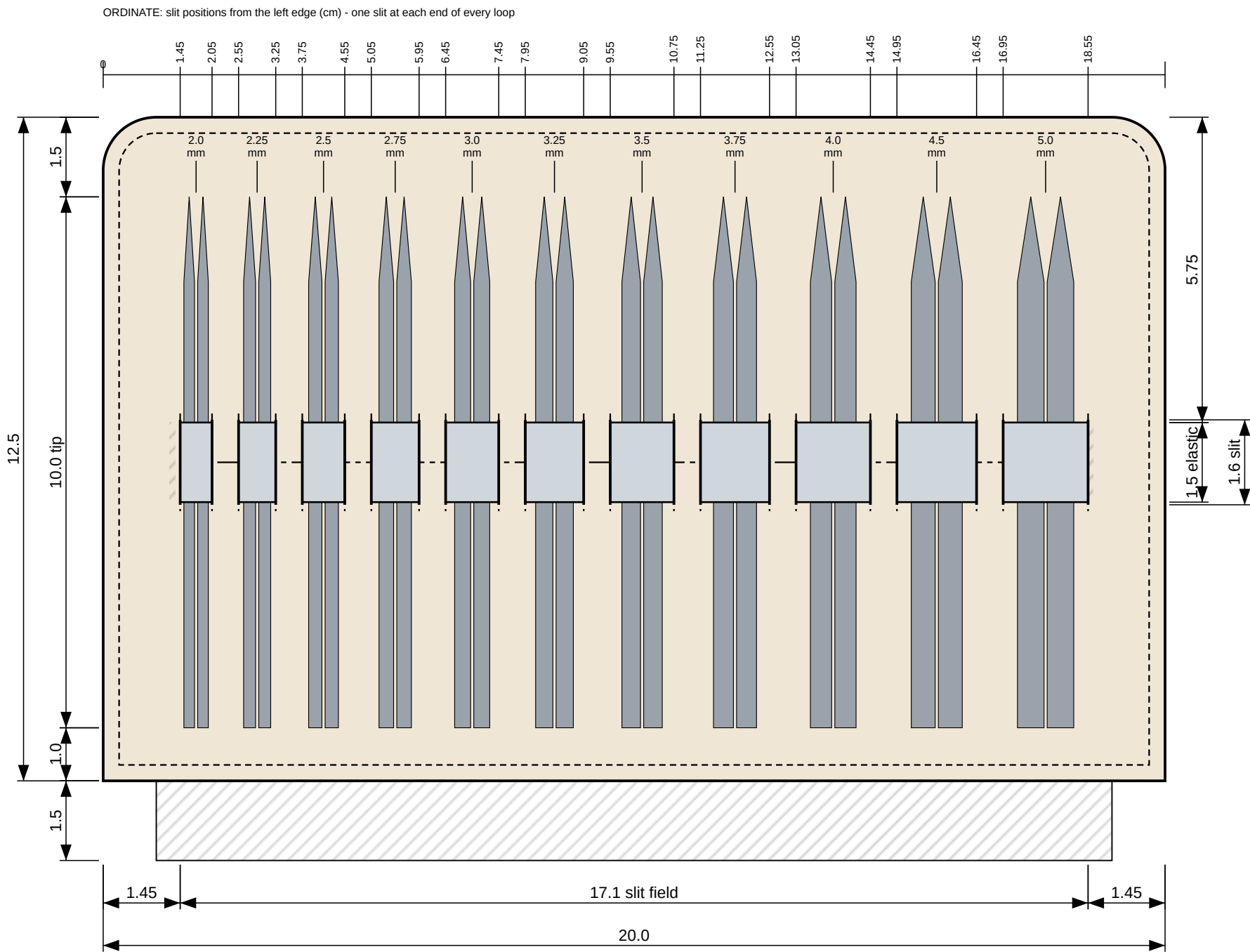


OPEN LAYOUT - inner face up, flaps flat (1:4)



PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Assembly - closed views, section, open layout, closing sequence	1:2 / 1:1 / 1:4 as noted	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	2026-10-01 Rev E	3 of 9

SHEET 4 - PANEL 1 (PAGE): 10 CM TIPS 2.0-5.0 MM (11 PAIRS)



PANEL 1 (PAGE) - page 20.0 x 12.5 + 1.5 hinge tab. Full size 1:1 - use as slit template.

LOOP TABLE - panel 1 (slit positions from the left edge, cm)

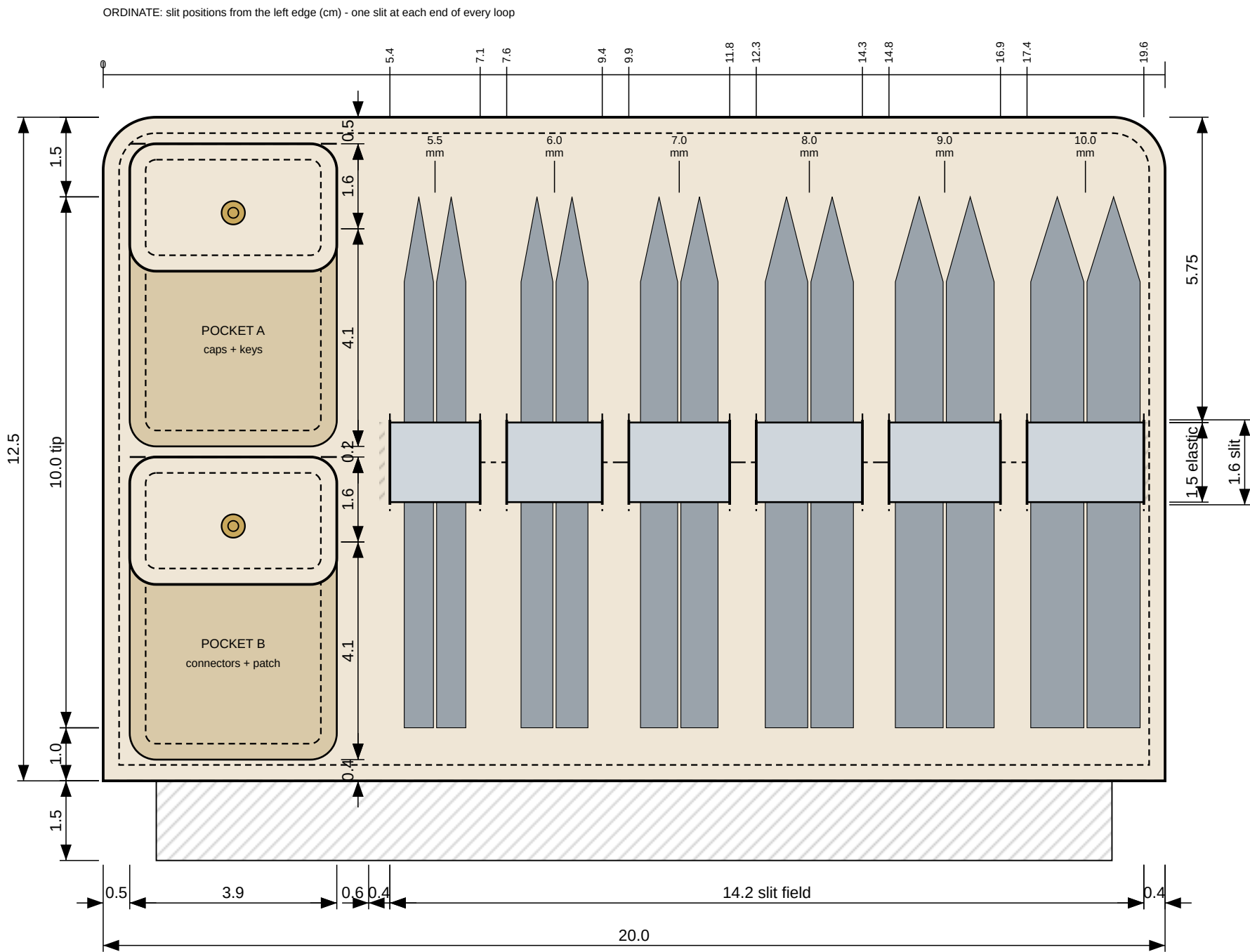
#	Size	Slot w	From	To	Holds
1	2.0 mm	0.6	1.45	2.05	2 x 10.0 cm
2	2.25 mm	0.7	2.55	3.25	2 x 10.0 cm
3	2.5 mm	0.8	3.75	4.55	2 x 10.0 cm
4	2.75 mm	0.9	5.05	5.95	2 x 10.0 cm
5	3.0 mm	1.0	6.45	7.45	2 x 10.0 cm
6	3.25 mm	1.1	7.95	9.05	2 x 10.0 cm
7	3.5 mm	1.2	9.55	10.75	2 x 10.0 cm
8	3.75 mm	1.3	11.25	12.55	2 x 10.0 cm
9	4.0 mm	1.4	13.05	14.45	2 x 10.0 cm
10	4.5 mm	1.5	14.95	16.45	2 x 10.0 cm
11	5.0 mm	1.6	16.95	18.55	2 x 10.0 cm

PANEL CONSTRUCTION (all four storage surfaces)

- Face ply 1.0 mm leather with the slits die-cut, over 0.8 mm board and a 0.6 mm back ply; edge-stitched 0.3 in contrast thread. Pages: R 1.0 top corners.
- Elastic: 15 mm knit elastic, matching colour, threaded IN and OUT of the face ply through 1.6 slits so each pair sits in its own visible loop and the elastic runs hidden behind the face between loops (reference photos).
- Loop (slot) widths are the prototype's: 0.6 for 2.0 mm, +0.1 per size step, to 2.2 for 10 mm. 0.5 hidden run between loops. One loop holds one PAIR, side by side.
- Thread the elastic before laminating the face to the board. Bar-tack 1.5 at every slit; stitch the two ends down under a 1.0 anchor. Elastic cut = slit field + 2.
- Size labels heat-stamped / printed above each loop with a 0.5 leader line to the tip (reference photo).
- Tip points up; 1.5 clear above the points (prototype).

PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Panel 1 (page): 10 cm tips 2.0-5.0 mm (11 pairs)	1:1	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	2026-10-01 Rev E	4 of 9

SHEET 5 - PANEL 2 (PAGE): 10 CM TIPS 5.5-10 MM + ACCESSORY POCKETS



PANEL 2 - page 20.0 x 12.5 + 1.5 hinge tab. Full size 1:1 - use as slit template.

LOOP TABLE - panel 2 (slit positions from the left edge, cm)

#	Size	Slot w	From	To	Holds
1	5.5 mm	1.7	5.4	7.1	2 x 10 cm
2	6.0 mm	1.8	7.6	9.4	2 x 10 cm
3	7.0 mm	1.9	9.9	11.8	2 x 10 cm
4	8.0 mm	2.0	12.3	14.3	2 x 10 cm
5	9.0 mm	2.1	14.8	16.9	2 x 10 cm
6	10.0 mm	2.2	17.4	19.6	2 x 10 cm

PANEL 2 LAYOUT (reference photo R5)

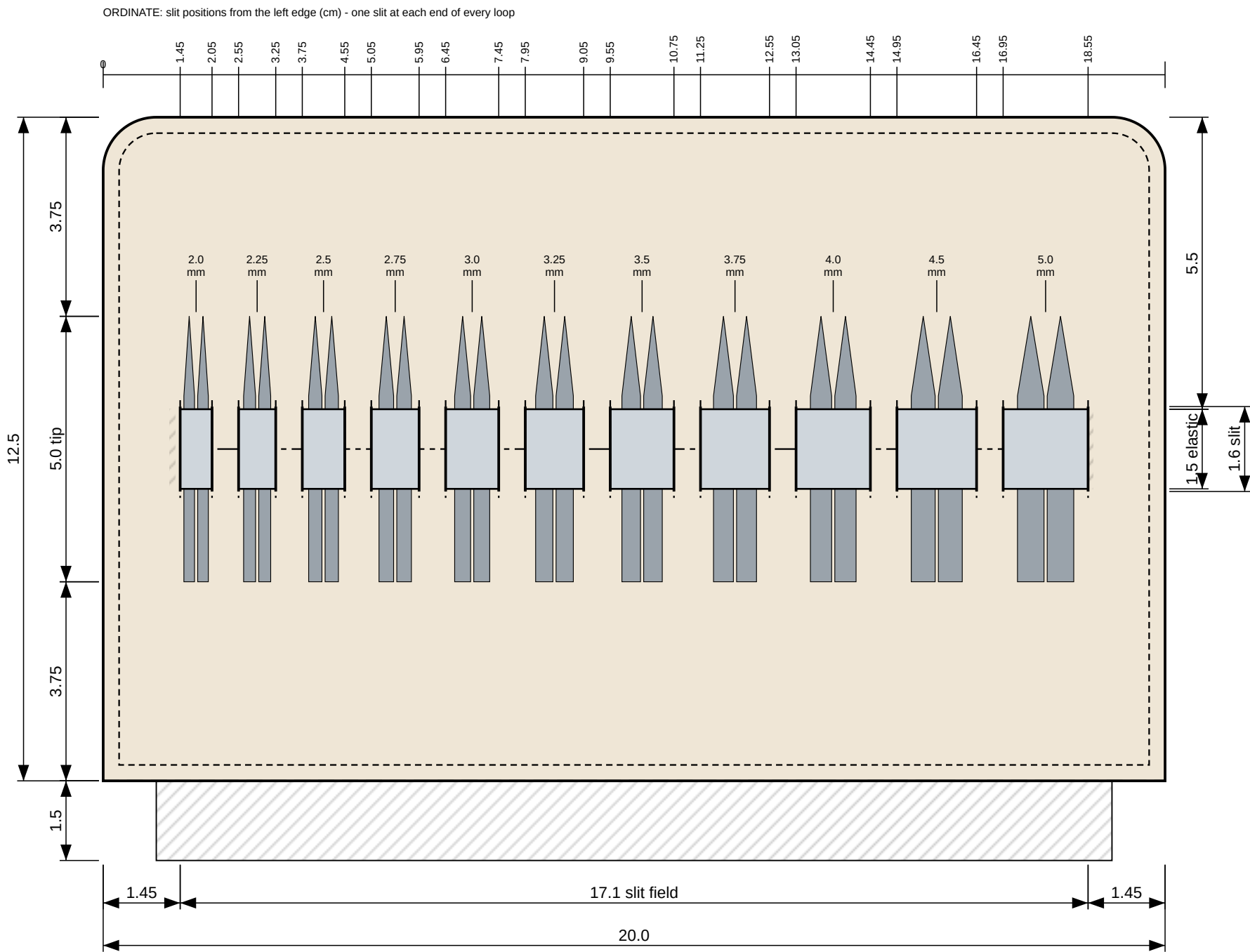
- Two small patch pockets at the left, stacked, as large as the page allows beside the 15.0 needle zone: body 3.9 x 4.1, flap 3.9 x 1.6 with a 10 mm brass snap, edge-stitched. Body cut 3.9 x 4.6 (0.5 under the flap). Pocket A: 6 end caps + 4 keys. Pocket B: 3 connectors + the folded leather grip patch (6.0 x 3.0, folds to 3.0).
- Needle zone 15.0 wide: 6 pairs 5.5-10 mm, loop widths 1.7-2.2, slit field 14.2 centred (0.4 margins).
- The 10 mm pair is the thickest item in the case (1.0).

PANEL CONSTRUCTION (all four storage surfaces)

1. Face ply 1.0 mm leather with the slits die-cut, over 0.8 mm board and a 0.6 mm back ply; edge-stitched 0.3 in contrast thread. Pages: R 1.0 top corners.
2. Elastic: 15 mm knit elastic, matching colour, threaded IN and OUT of the face ply through 1.6 slits so each pair sits in its own visible loop and the elastic runs hidden behind the face between loops (reference photos).
3. Loop (slot) widths are the prototype's: 0.6 for 2.0 mm, +0.1 per size step, to 2.2 for 10 mm. 0.5 hidden run between loops. One loop holds one PAIR, side by side.
4. Thread the elastic before laminating the face to the board. Bar-tack 1.5 at every slit; stitch the two ends down under a 1.0 anchor. Elastic cut = slit field + 2.
5. Size labels heat-stamped / printed above each loop with a 0.5 leader line to the tip (reference photo).
6. Tip points up; 1.5 clear above the points (prototype).

PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Panel 2 (page): 10 cm tips 5.5-10 mm + accessory pockets	1:1	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	2026-10-01 Rev E	5 of 9

SHEET 6 - PANEL 3 (PAGE): 5 CM TIPS 2.0-5.0 MM (11 PAIRS)



PANEL 3 (PAGE) - page 20.0 x 12.5 + 1.5 hinge tab. Full size 1:1 - use as slit template.

LOOP TABLE - panel 3 (slit positions from the left edge, cm)

#	Size	Slot w	From	To	Holds
1	2.0 mm	0.6	1.45	2.05	2 x 5.0 cm
2	2.25 mm	0.7	2.55	3.25	2 x 5.0 cm
3	2.5 mm	0.8	3.75	4.55	2 x 5.0 cm
4	2.75 mm	0.9	5.05	5.95	2 x 5.0 cm
5	3.0 mm	1.0	6.45	7.45	2 x 5.0 cm
6	3.25 mm	1.1	7.95	9.05	2 x 5.0 cm
7	3.5 mm	1.2	9.55	10.75	2 x 5.0 cm
8	3.75 mm	1.3	11.25	12.55	2 x 5.0 cm
9	4.0 mm	1.4	13.05	14.45	2 x 5.0 cm
10	4.5 mm	1.5	14.95	16.45	2 x 5.0 cm
11	5.0 mm	1.6	16.95	18.55	2 x 5.0 cm

PANEL CONSTRUCTION (all four storage surfaces)

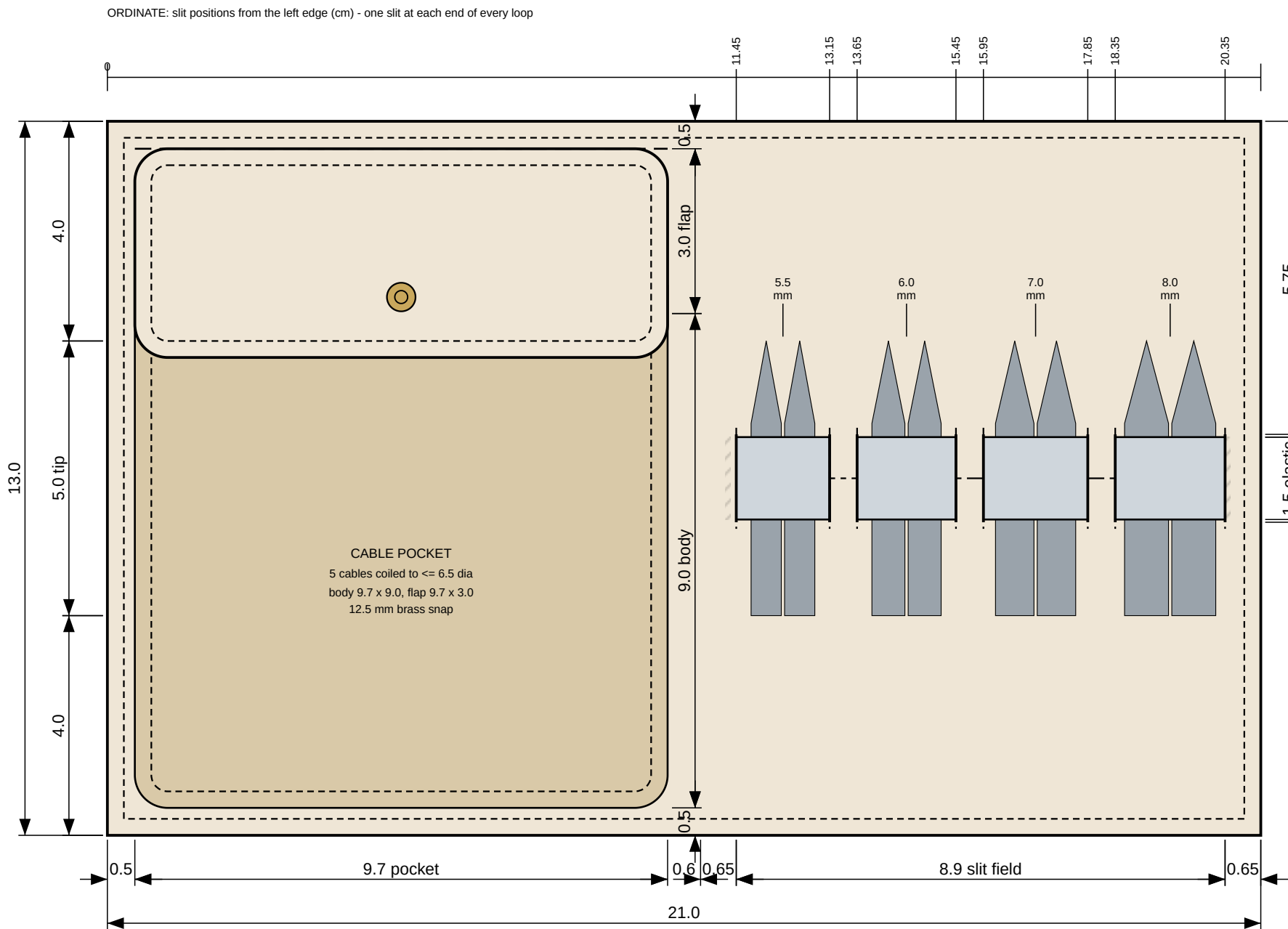
1. Face ply 1.0 mm leather with the slits die-cut, over 0.8 mm board and a 0.6 mm back ply; edge-stitched 0.3 in contrast thread. Pages: R 1.0 top corners.
2. Elastic: 15 mm knit elastic, matching colour, threaded IN and OUT of the face ply through 1.6 slits so each pair sits in its own visible loop and the elastic runs hidden behind the face between loops (reference photos).
3. Loop (slot) widths are the prototype's: 0.6 for 2.0 mm, +0.1 per size step, to 2.2 for 10 mm. 0.5 hidden run between loops. One loop holds one PAIR, side by side.
4. Thread the elastic before laminating the face to the board. Bar-tack 1.5 at every slit; stitch the two ends down under a 1.0 anchor. Elastic cut = slit field + 2.
5. Size labels heat-stamped / printed above each loop with a 0.5 leader line to the tip (reference photo).
6. Tip points up; 1.5 clear above the points (prototype).

PANEL 3 NOTE

- Same loop widths and slit positions as panel 1, so the two pages share one slit-cutting template.
- 5 cm tips sit centred on the page (reference photo): points 3.75 from the top, elastic centred at 6.25.

PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Panel 3 (page): 5 cm tips 2.0-5.0 mm (11 pairs)	SCALE 1:1	UNITS cm (page A3)
SOURCE Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	DATE / REV 2026-10-01 Rev E	SHEET 6 of 9

SHEET 7 - PANEL 4 (BACK WALL): 5 CM TIPS 5.5-8 MM + CABLE POCKET



LOOP TABLE - panel 4 (slit positions from the left edge, cm)

#	Size	Slot w	From	To	Holds
1	5.5 mm	1.7	11.45	13.15	2 x 5 cm
2	6.0 mm	1.8	13.65	15.45	2 x 5 cm
3	7.0 mm	1.9	15.95	17.85	2 x 5 cm
4	8.0 mm	2.0	18.35	20.35	2 x 5 cm

PANEL 4 - BACK WALL (reference photo R6)

- This is the lining ply of the back wall, 21.0 x 13.0: the loops and the pocket are made on it BEFORE it is laminated to the board and outer shell, so the hidden elastic runs lie between lining and board.
- Cable pocket at the left, as large as the wall allows beside the needle zone: patch pocket 9.7 x 9.0 with a 3.0 flap and 12.5 mm brass snap. Body cut 9.7 x 9.5. Holds the 5 cables, each coiled to 6.5 cm or less.
- Needle zone 10.2 wide: 4 pairs 5.5 / 6 / 7 / 8 mm, loop widths 1.7-2.0, slit field 8.9 centred (0.65 margins).
- 5 cm tips: points 4.0 from the top, elastic centred at 6.5 (level with panels 1-2).

PANEL CONSTRUCTION (all four storage surfaces)

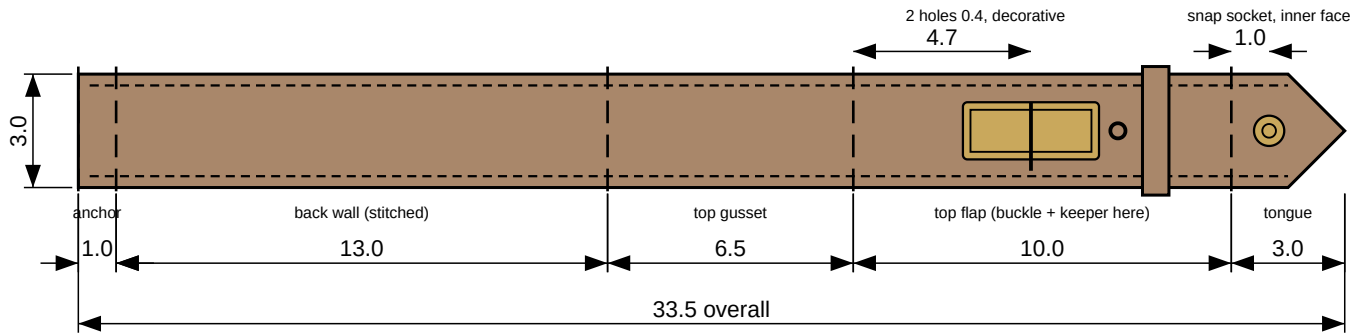
1. Face ply 1.0 mm leather with the slits die-cut, over 0.8 mm board and a 0.6 mm back ply; edge-stitched 0.3 in contrast thread. Pages: R 1.0 top corners.
2. Elastic: 15 mm knit elastic, matching colour, threaded IN and OUT of the face ply through 1.6 slits so each pair sits in its own visible loop and the elastic runs hidden behind the face between loops (reference photos).
3. Loop (slot) widths are the prototype's: 0.6 for 2.0 mm, +0.1 per size step, to 2.2 for 10 mm. 0.5 hidden run between loops. One loop holds one PAIR, side by side.
4. Thread the elastic before laminating the face to the board. Bar-tack 1.5 at every slit; stitch the two ends down under a 1.0 anchor. Elastic cut = slit field + 2.
5. Size labels heat-stamped / printed above each loop with a 0.5 leader line to the tip (reference photo).
6. Tip points up; 1.5 clear above the points (prototype).

PANEL 4 - BACK-WALL LINING 21.0 x 13.0 (inner face of the back wall; not a loose page). Full size 1:1 - use as slit template.

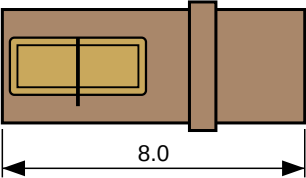
PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Panel 4 (back wall): 5 cm tips 5.5-8 mm + cable pocket	SCALE 1:1	UNITS cm (page A3)
SOURCE Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	DATE / REV 2026-10-01 Rev E	SHEET 7 of 9

SHEET 8 - DETAILS - STRAP, ELASTIC THREADING, PAGE HINGE, POCKETS

DETAIL B - STRAP, flat, 1:2 (dark brown leather 1.4 mm, 2 ply, edge-stitched 0.3)



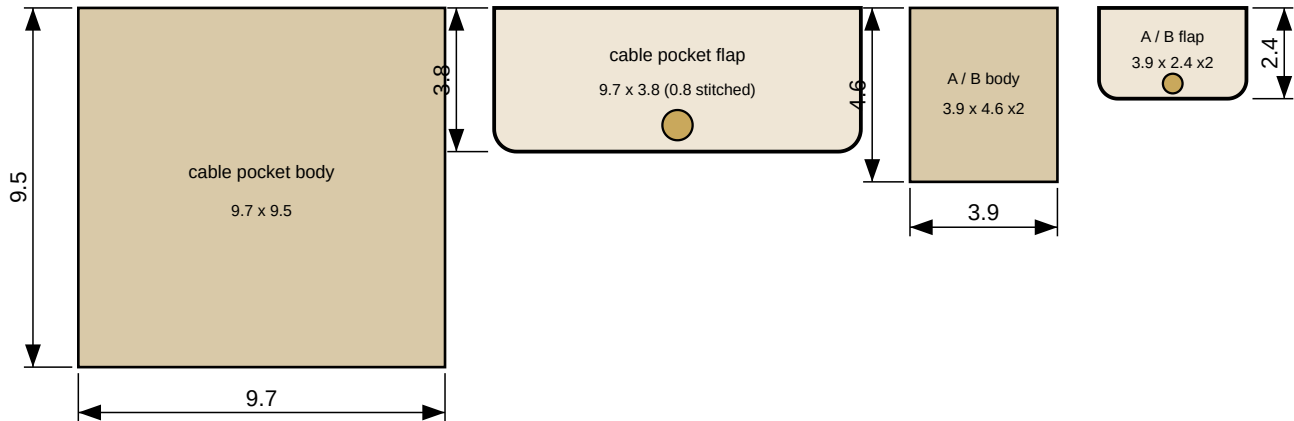
BUCKLE PIECE - 3.0 x 8.0 loop round the 30 mm buckle bar, stitched ON the strap over the top flap (bar 4.7 below the flap top edge); keeper 3.4 x 0.7 stitched at 8.0. Both decorative.



Buckle: 30 mm (1 1/4 in) antique brass, fixed; the tongue passes under the keeper and closes with the 12.5 mm snap.

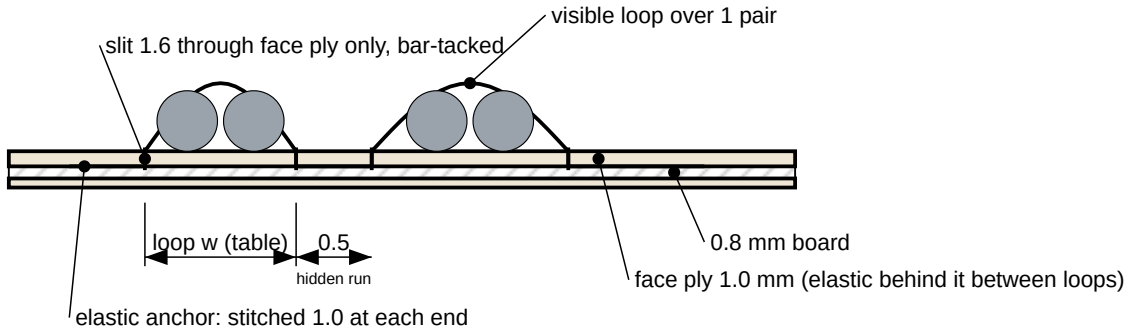
Stud on the front panel outer face, centred, 11.0 below the top edge (sheet 2). Set after a dry fold.

DETAIL D - POCKET PATTERNS, 1:2 (net; add 0.8 turn-under on the three stitched sides, top edge bound or skived)

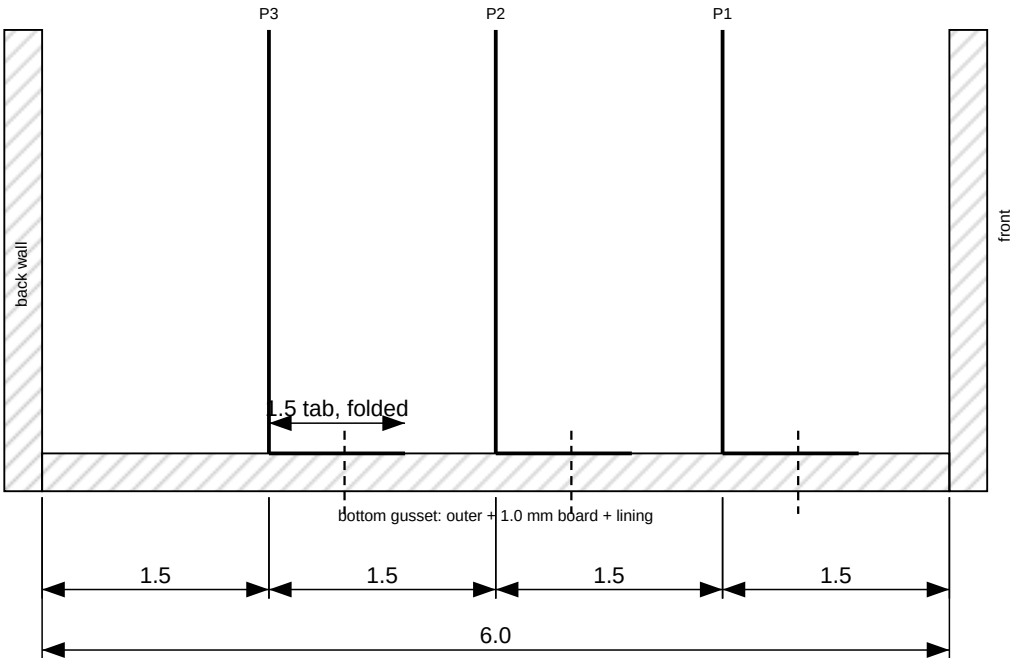


- Pockets are patch pockets: body stitched to the panel face on three sides 0.3 from the edge, over a 0.8 turn-under; flap stitched along its top edge 0.5 in from the fold. Snaps: 12.5 mm (cable pocket), 10 mm (A, B).
- Make the pockets on the face ply before the elastic is threaded and the ply laminated.

DETAIL A - elastic threaded through the face ply, section, 2:1



DETAIL C - PAGE HINGES, section through the bottom gusset, 2:1



- Each page's 1.5 hinge tab folds toward the front and is stitched through the gusset (lining + board + outer) on one line, 0.5 from the fold. Three seams evenly spaced at 1.5 so the pages fan open like a book (photo R2).
- Stitch P3, then P2, then P1 from the back; the gusset outer face shows three parallel stitch lines - use the same contrast thread. Pages are not removable.

PROJECT	CLIENT	
INTERCHANGEABLE KNITTING NEEDLE CASE	gsumano7 / ziqmac	
SHEET TITLE	SCALE	UNITS
Details - strap, elastic threading, page hinge, pockets	1:2 / 2:1 as noted	cm (page A3)
SOURCE	DATE / REV	SHEET
Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	2026-10-01 Rev E	8 of 9

SHEET 9 - CONTENTS CHECKLIST, BILL OF MATERIALS, CONSTRUCTION & CUTTING LIST

CONTENTS CHECKLIST - where each item lives

Size	10 cm tips	Panel	Loop	w	5 cm tips	Panel	Loop	w
2.0 mm	2 x 10 cm	Panel 1	1	0.6	2 x 5 cm	Panel 3	1	0.6
2.25 mm	2 x 10 cm	Panel 1	2	0.7	2 x 5 cm	Panel 3	2	0.7
2.5 mm	2 x 10 cm	Panel 1	3	0.8	2 x 5 cm	Panel 3	3	0.8
2.75 mm	2 x 10 cm	Panel 1	4	0.9	2 x 5 cm	Panel 3	4	0.9
3.0 mm	2 x 10 cm	Panel 1	5	1.0	2 x 5 cm	Panel 3	5	1.0
3.25 mm	2 x 10 cm	Panel 1	6	1.1	2 x 5 cm	Panel 3	6	1.1
3.5 mm	2 x 10 cm	Panel 1	7	1.2	2 x 5 cm	Panel 3	7	1.2
3.75 mm	2 x 10 cm	Panel 1	8	1.3	2 x 5 cm	Panel 3	8	1.3
4.0 mm	2 x 10 cm	Panel 1	9	1.4	2 x 5 cm	Panel 3	9	1.4
4.5 mm	2 x 10 cm	Panel 1	10	1.5	2 x 5 cm	Panel 3	10	1.5
5.0 mm	2 x 10 cm	Panel 1	11	1.6	2 x 5 cm	Panel 3	11	1.6
5.5 mm	2 x 10 cm	Panel 2	1	1.7	2 x 5 cm	Panel 4	1	1.7
6.0 mm	2 x 10 cm	Panel 2	2	1.8	2 x 5 cm	Panel 4	2	1.8
7.0 mm	2 x 10 cm	Panel 2	3	1.9	2 x 5 cm	Panel 4	3	1.9
8.0 mm	2 x 10 cm	Panel 2	4	2.0	2 x 5 cm	Panel 4	4	2.0
9.0 mm	2 x 10 cm	Panel 2	5	2.1	-	-	-	-
10.0 mm	2 x 10 cm	Panel 2	6	2.2	-	-	-	-

Totals: 17 pairs x 10 cm (34 tips) on panels 1 + 2;
15 pairs x 5 cm (30 tips) on panels 3 + 4; 32 pairs / 64 tips.
Accessories: 6 caps + 4 keys -> panel 2 pocket A;
3 connectors + grip patch -> pocket B; 5 cables -> panel 4 pocket.

BILL OF MATERIALS (per case)

#	Item	Qty	Cut size (cm) / note
1	Outer shell, full-grain leather 1.2-1.4 mm (or PU 1.0-1.2), tan	1	48.4 x 48.2 net + allowance (sheet 2)
2	Lining 0.6 mm, tan (back-wall lining = panel 4, sheet 7)	1	48.4 x 48.2 net + allowance
3	Board 1.0 mm: back wall, front panel	2	21 x 13, 21 x 12.7
4	Board 0.8 mm: pages	3	20.0 x 12.5
5	Page face plies 1.0 mm (slits die-cut) + back plies 0.6 mm	3+3	20.0 x 14.0 (incl. 1.5 hinge tab) + allowance
6	Knit elastic 15 mm, tan	4	19.1 / 16.2 / 19.1 / 10.9 cm (slit field + 2)
7	Strap, leather 1.4 mm, dark brown, 2 ply	2	3.0 x 33.5 (sheet 8)
8	Buckle piece + keeper, dark brown	1+1	3.0 x 8.0; 3.4 x 0.7
9	Buckle 30 mm, antique brass	1	decorative, fixed on the strap
10	Spring snaps 12.5 mm (line 20), antique brass	4 sets	2 side flaps, 1 strap tongue, 1 cable pocket
11	Spring snaps 10 mm, antique brass	2 sets	pockets A, B
12	Cable pocket body + flap, 0.8 mm leather	1+1	9.7 x 9.5; 9.7 x 3.8
13	Small pocket bodies + flaps, 0.8 mm leather	2+2	3.9 x 4.6; 3.9 x 2.4
14	Leather grip patch, veg-tan 1.8-2 mm	1	6.0 x 3.0, R 0.5, folds in half
15	Thread, bonded polyester Tex 70, contrast	-	stitch length 3 mm
16	Logo emboss die	1	2.5 x 2.5 (client artwork, sheet 1)

CONSTRUCTION SEQUENCE

- Cut shell, lining, board, pages and pockets (sheets 2, 4-8). Die-cut the elastic slits in the three page face plies and in the back-wall lining (sheets 4-7 at 1:1). Emboss the logo on the top flap while flat.
- Make the pockets on the face plies (panel 2: A, B; back-wall lining: cable pocket). Set the pocket snaps.
- Thread the elastics in and out of the slits, loop by loop; bar-tack each slit, stitch the ends down. Print / stamp the size labels.
- Laminate each page: face ply + 0.8 board + back ply; edge-stitch 0.3, leaving the 1.5 hinge tab single-ply.
- Stitch the strap to the outer shell (back wall + top gusset + top flap) with the buckle piece and keeper on the flap section.
- Set the side-flap studs and the front-panel sockets. Laminate lining to shell with board in the back wall and front panel.
- Stitch the page hinge tabs through the bottom gusset in order P3, P2, P1 (sheet 8, detail C). Turn / bind all edges; edge-stitch 0.3.
- Dry-fold; set the strap stud on the front panel and the socket in the tongue; check all snaps. Load the set; the closed case must not press on the 10 mm tips.

TOLERANCES

Cut pieces +/- 0.1 cm; slit positions +/- 0.05 cm (the 0.6 loop must still take two 2 mm tips); snap centres +/- 0.15 cm; hinge seams +/- 0.1 cm.

MEASUREMENT RECORD - what each prototype / reference photo contributed

Photo	Feature	Value used on drawings
P1	Cardboard cross pattern	Top gusset 6.5, back 21.0 x 13.0, bottom gusset 6.0, front 12.7; side gusset 5.7, side flap 8.0, snaps. Top flap 5.0 superseded by 10.0 (client, per R1/R3)
P2-P5	Cardboard panels	Loop widths 0.6-2.2 (= 6-22 mm), 0.5 between loops, 1.0 end margins, 1.5 above the points
R1, R3	Closed clutch, front/back	Trifold clutch, wrap strap 3.0 with buckle, keeper; logo on flap; R 1.0 corners; contrast edge stitch
R2, R3	Open case	Pages sewn into the bottom gusset, not removable; side flaps tuck inside
R4-R6	Panel interiors	1.5 elastic threaded in / out of the panel per pair; size labels above tips; snap pockets on panel 2 and the back wall
Brief	Client text 2026-10-01	Front > P1 > P2 > P3 > back wall; back wall = panel 4 with cable pocket; 2 small pockets on panel 2; snaps hold the front panel to the side flaps; strap visual, snap-closed; logo

HARDWARE SIZE ASSUMPTIONS (confirm against the real set - Q5)

Item	Qty	Assumed size (cm)	Where
Screw-on end cap	6	1.0 dia x 1.5	panel 2 pocket A
Cable key	4	1.2 x 3.5	panel 2 pocket A
Cable connector	3	1.0 dia x 3.0	panel 2 pocket B
Leather grip patch	1	6.0 x 3.0 x 0.2, folded	panel 2 pocket B
Cables 25-100 cm	5	coil to 6.5 dia	panel 4 cable pocket

PROJECT INTERCHANGEABLE KNITTING NEEDLE CASE	CLIENT gsumano7 / ziqmac	
SHEET TITLE Contents checklist, bill of materials, construction & cutting list	SCALE -	UNITS cm (page A3)
SOURCE Cardboard prototype P1-P5 + client brief, logo and reference photos R1-R6	DATE / REV 2026-10-01 Rev E	SHEET 9 of 9